BIOT Module 3

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1. Introduction

In Module 2 I ended up with a ranked list of 15 Alzheimer's genes. But a gene being linked to a disease does not mean a drug can act on it. Some proteins have no place for a drug to bind, and some sit inside the cell where an antibody cannot reach.

In this module I used protein language models, which turn each protein sequence into a list of numbers, to compare my 15 proteins with proteins that already have drugs. If one of my proteins looks like proteins that have been drugged before, that is a good sign.

This brings us to the main question of this module: which of my 15 Alzheimer's targets look like proteins a drug could act on, and should that drug be a small molecule or an antibody?

2. Disease and data choice, with biological interpretation

I stayed with Alzheimer's disease (AD), carried over from Module 2. Its genetics point to brain immune cells, lipid handling and synapses, and my 15 targets come from all three.

The data is the course data pack. It has ESM C 300M and ESM-2 embeddings for 20,190 reviewed human proteins, a 1,201-protein training panel labelled with Open Targets clinical precedence, and ESM-2 650M scores for the anchor variants. Every protein gets the same kind of representation, so my 15 targets are scored on equal terms against proteins that have already been drugged.

What the re-ranked targets mean:

- *The top four are surface proteins of immune cells and blood vessels* (PECAM1, HLA-DRB1, HLA-DRA, ITGAX). This fits AD biology: microglia and the blood–brain barrier change early in the disease, and antibodies can reach proteins on the cell surface.
- *Known AD genes are recovered, but tractability splits them:* HLA-DRB1 stays 2nd, while APOE, the strongest genetic risk gene, falls to 13th because it resembles no drugged protein.
- *Surprise, APLNR:* the apelin receptor is not a classic AD gene, yet it rises four places to 6th. It is a GPCR, so it becomes my best small-molecule candidate.

Key terms: p_sm and p_ab are the model's probabilities that a protein is tractable by a small molecule or by an antibody, learned from proteins that already have drugs.

3. Re-ranking

Did the prior agree with my Module 2 ranking? Mostly not, and that is expected. The two scores measure different things (disease link against resemblance to drugged proteins), and their Spearman correlation is only 0.21. The small positive value comes from the antibody side (0.38 with p_ab, −0.16 with p_sm), because my strongest Module 2 hits are immune and cell-surface proteins.

- *Looks druggable:* PECAM1 (p_ab 0.88), APLNR (p_sm 0.68), ITGAX (p_ab 0.59), MOG (p_ab 0.50).
- *In between (0.1 to 0.5):* NRXN3, SHANK2, HLA-DRB1, APOC1, HLA-DRA, GMPR, APOB, MTSS2.
- *No precedent for either drug type (below 0.1):* RPH3A, APOE, PLEC.

Table 1: Final ranking (w = 0.6). APOE and APOB tie on the final score (0.253); the tie is broken by Module 2 order.

Final	Gene	Module 2	Change	p_sm	p_ab	Nearest drugged proteins (cosine)
1	PECAM1	4	+3	0.02	0.88	CD2 (0.96), IL12B (0.96), LAMA4 (0.95)
2	HLA-DRB1	2	0	0.01	0.27	SIGLEC8 (0.87), IFNA10 (0.86), PTGDS (0.85)
3	HLA-DRA	1	−2	0.01	0.18	SEMA4D (0.92), GRIK3 (0.92), CA12 (0.91)
4	ITGAX	6	+2	0.21	0.59	ITGA2B (0.98), CSF1R (0.96), SEMA4D (0.95)
5	APOC1	3	−2	0.00	0.25	IFNA5 (0.86), SCTR (0.85), SLC34A2 (0.85)
6	APLNR	10	+4	0.68	0.03	AGTR2 (0.96), GHSR (0.95), S1PR2 (0.94)
7	NRXN3	9	+2	0.44	0.12	SDC4 (0.94), KCNB1 (0.93), RELA (0.92)
8	MTSS2	5	−3	0.12	0.01	PDE4D (0.94), KCNQ4 (0.93), PDE4C (0.93)
9	GMPR	7	−2	0.17	0.00	EEF2 (0.95), PSMA5 (0.93), RPL11 (0.93)
10	SHANK2	12	+2	0.33	0.00	E2F7 (0.97), MAP3K13 (0.96), KCNQ3 (0.95)
11	MOG	14	+3	0.00	0.50	PDCD1LG2 (0.93), TIGIT (0.92), IFNGR2 (0.91)
12	RPH3A	8	−4	0.09	0.00	CACNB3 (0.93), PDE4D (0.93), CACNB2 (0.92)
13	APOE	11	−2	0.00	0.01	MADCAM1 (0.86), TNFSF14 (0.85), IL11RA (0.84)
14	APOB	13	−1	0.15	0.10	AFP (0.97), LAMA4 (0.95), IFNAR2 (0.95)
15	PLEC	15	0	0.01	0.00	LAMB2 (0.89), EPHA10 (0.87), EPHA1 (0.86)

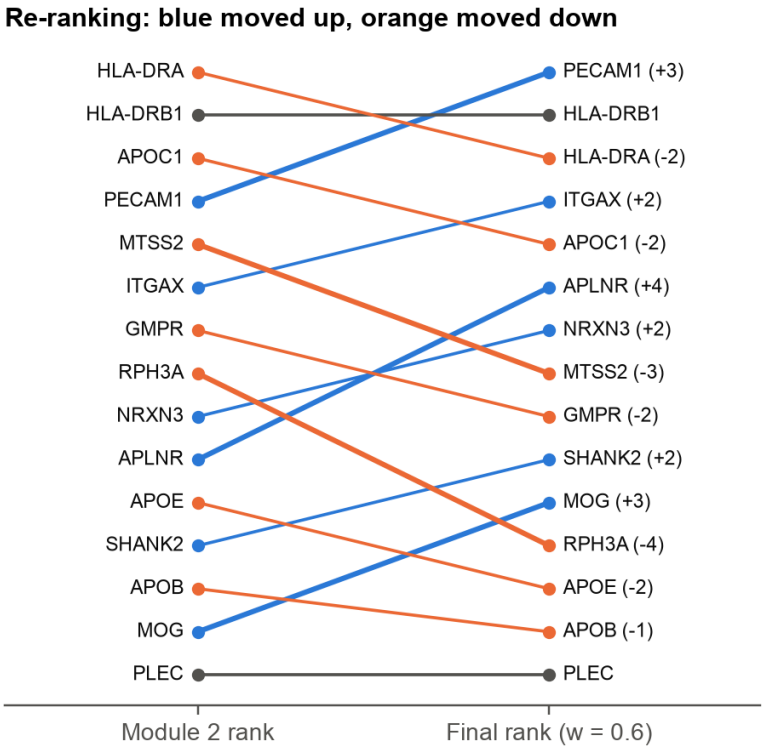


Figure 1: How each target moved between the Module 2 ranking and the final ranking.

Which targets moved most, and why? Two moved four places:

- *RPH3A fell furthest (8th to 12th):* a synaptic-vesicle protein inside the cell that works by binding other proteins and membranes, not through a drug-like pocket. Antibodies cannot reach it (p_ab 0.00) and its small-molecule prior is low (0.09). Its nearest "drugged" proteins, CACNB3 and CACNB2, are probably labelled drugged only because the calcium channel they belong to is a drug target.
- *APLNR rose furthest (10th to 6th):* a GPCR whose nearest drugged proteins (AGTR2, GHSR, S1PR2) are GPCRs of the same class, the classic small-molecule family.

PECAM1 and MOG also rose three places each, on strong antibody priors.

Why w = 0.6: I weighted Module 2 evidence slightly above tractability.

1. *Disease relevance comes first:* a druggable protein that does not cause AD is a wasted programme. Tractability can be worked around (another drug type, or a partner in the same pathway); causality cannot.
2. *The prior is only moderately reliable:* its family-split AUROC is 0.745 (Section 5), so it should adjust the ranking, not override it.
3. *Sensitivity:* at w = 0.5 the prior starts to override strong genetics (HLA-DRA, the Module 2 #1, falls to 4th behind ITGAX). At w = 0.7 the top six are exactly my Module 2 top six, and APLNR, my best small-molecule target, drops to 7th.

4. Modality

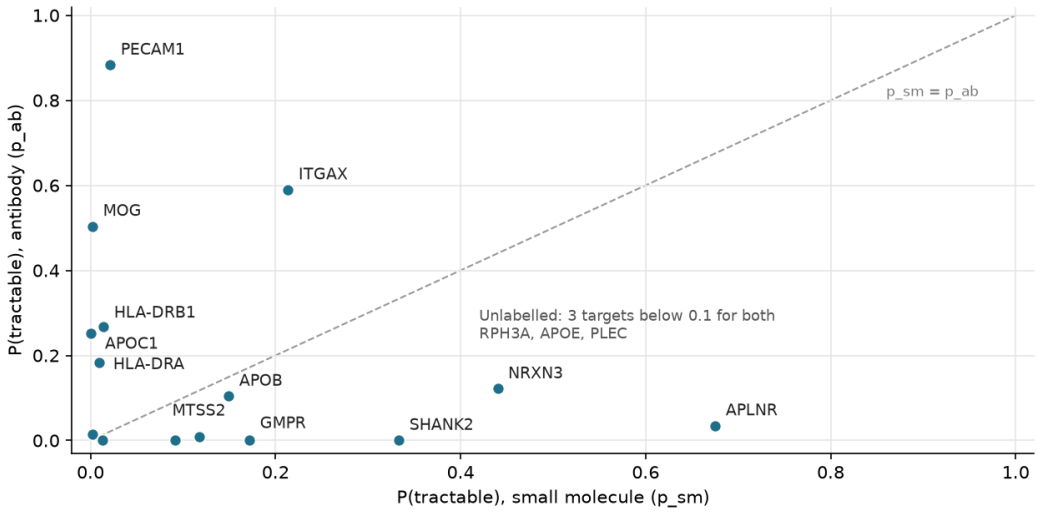


Figure 2: Small-molecule against antibody prior for my 15 targets. Cell-surface targets (PECAM1, ITGAX, MOG) sit high on the antibody axis; only the GPCR APLNR sits far out on the small-molecule axis.

Target: PECAM1 (p_ab 0.88, p_sm 0.02), the sharpest antibody-over-small-molecule split.

- *Location:* a membrane protein on blood-vessel lining cells, platelets and white blood cells, with a large outer part exposed to the blood.
- *Family:* six antibody-like (immunoglobulin) domains. Its nearest drugged proteins (CD2, IL12B, LAMA4) all work outside the cell, and CD2 and IL12B are antibody-drugged.
- *Likely pocket:* none deep. These domains bind partners through flat, broad surfaces that small molecules cannot grip.
- *AD angle:* PECAM1 lines the blood vessels of the blood–brain barrier, so an antibody can reach it from the blood without crossing into the brain.

The mirror image, APLNR (p_sm 0.68, p_ab 0.03): a GPCR that crosses the membrane seven times. Its natural ligand binds in a pocket inside this bundle, which is where most GPCR drugs bind. The part outside the cell that an antibody could reach is small. The panel has 53 GPCRs of this class, 33 of them with clinical precedence, so this prior rests on a well-populated family.

The hard case, APOE (p_sm 0.00, p_ab 0.01): a secreted lipid carrier with no enzyme pocket or receptor shape, so it looks like neither drug class. That is a drug-type problem, not a reason to drop it. Options include small molecules that correct the shape of ApoE4, antibodies against ApoE stuck in plaques, or targeting its partners (LDLR, ABCA1).

5. Validation

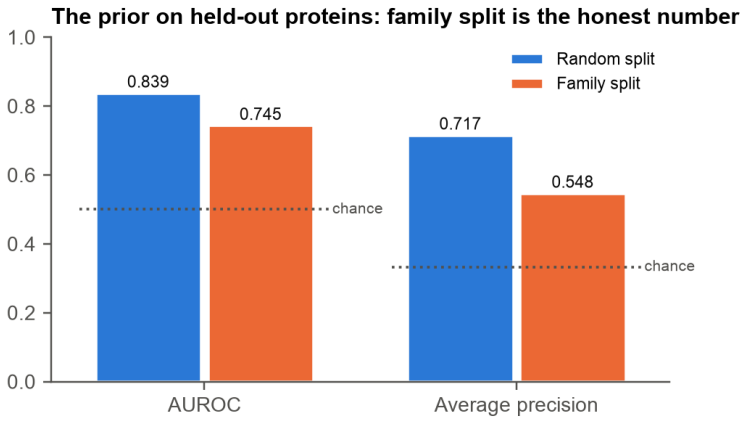


Figure 3: The same model tested two ways. In the family split, whole protein families are held out together.

Why the family split is the honest number: in a random split, close relatives sit on both sides. If AKT1 is in training, AKT3 is easy to "predict", which rewards memorising families rather than learning what makes a protein tractable. Here the family split is strict: all 61 kinases form one group and all 53 GPCRs of the main class another. That matches a new AD target whose family may never have been drugged (APOC1, MTSS2, RPH3A).

What the random split would have hidden: a drop of 0.094 AUROC and 0.169 average precision. The gain over chance in average precision shrinks from 0.384 to 0.215, so the random split hides about 44% of it. The prior still beats chance, but a team reading 0.84 would trust it more than it deserves.

6. Variants

Positions follow UniProt, which counts APOE's 18-residue signal peptide, so they are 18 higher than the usual ApoE names (R176C is the familiar R158C, ε2; C130R is C112R, ε4). The score is log P(mutant) − log P(wild type) from ESM-2 650M. The percentile ranks each variant among every possible substitution in that protein, so a low percentile means more surprising than most.

Table 2: Anchor variants. The median substitution scores −3.9 in APOE, −4.8 in TREM2 and −2.2 in TP53.

Gene	Variant (UniProt)	Meaning	Score	Percentile in protein
APOE	R176C (rs7412)	ε2, protective for AD	−8.56	11.8
APOE	C130R (rs429358)	ε4, strongest common AD risk allele	+7.21	100.0
TREM2	R47H (rs75932628)	rare AD risk, weakens ligand binding	−5.16	47.7
TP53	R175H	cancer hotspot (breaks the fold)	−5.97	19.3

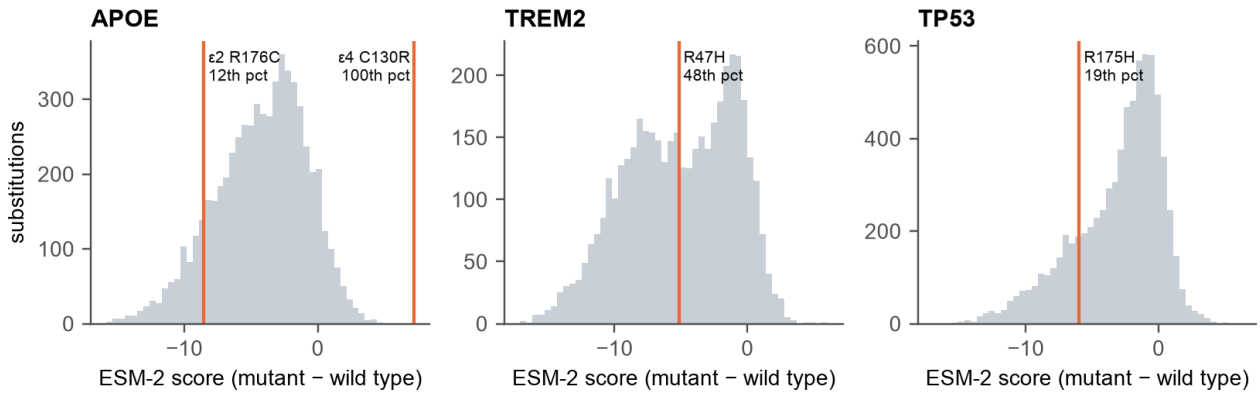


Figure 4: Every possible substitution in each protein (grey), with the anchor variants marked.

Reading the scores:

- *APOE ε4 (C130R) is the most "expected" change in all of APOE:* the model prefers arginine over the human cysteine here, because ε4 is the ancestral version and other mammals carry arginine. Yet it is the largest common genetic risk factor for AD.
- *APOE ε2 (R176C) is scored as damaging:* R176 sits where APOE binds the LDL receptor, and ε2 does weaken that binding. The model flags a real change that happens to protect against AD.
- *TREM2 R47H is middling:* no more surprising than a typical TREM2 change, despite raising AD risk two- to fourfold, because it subtly weakens binding on the surface rather than breaking the fold.
- *p53 R175H is in the damaging tail,* as expected for a hotspot that breaks the fold.

So the model ranks the two APOE alleles in the opposite order to their AD risk. A strongly negative score is not "pathogenic": it measures evolutionary surprise, not disease. A change can be surprising and harmless, or ordinary and disease-causing, like ε4.

What these models cannot see:

- *Variant types:* changes outside the protein code, splicing changes, insertions and deletions, and copy-number changes.
- *Context:* binding partners, cell type, expression level, one or two copies, age, and other variants.
- *Clinical evidence:* population frequency and the patient and family data that ClinVar uses.

7. Discordance, limitations and next step

A discordant gene, APOC1 (from Module 2): its RNA goes up in AD (+0.51) while its protein goes down (−0.23), and both changes are significant. Module 3 cannot say which layer is right: the language model reads only the protein sequence, which is the same either way, so APOC1 gets the same prior (p_ab 0.25) whichever layer is correct. The likely reasons are still those from Module 2: different brain regions, a secreted protein that leaves the tissue, or protein trapped in plaques.

The biggest disagreement between the two scores, APOE: the second-largest genetic signal in my list, but a prior near zero for both drug types. Both can be right at once. The prior says APOE does not look like proteins that have been drugged before; it does not say APOE is unimportant in AD.

What this lets me claim: which targets resemble proteins with clinical precedence, and for which drug type. *What it does not:* that any target is druggable, causes AD, or can be reached in the brain.

What sequence alone cannot tell me:

- *Pockets and 3D shape:* "looks like drugged proteins" is not "has a pocket a drug can bind", and hidden pockets (such as p53 Y220C) are invisible.
- *History in the labels:* labels record where industry has worked, so undrugged classes (apolipoproteins, scaffolds such as SHANK2 and PLEC) score low partly because nobody has tried. Labels can also pass to every part of a drugged complex, which is likely why two of GMPR's nearest "drugged" proteins are a proteasome subunit and a ribosomal protein.
- *An unbalanced antibody prior:* only 66 training proteins have antibody precedence, against 301 for small molecules.

- *Brain delivery:* nothing here models the blood–brain barrier, which decides whether a drug reaches targets on brain immune cells such as ITGAX or HLA-DR.
- *Direction and safety:* whether to block or activate, and side effects (HLA-DR is central to immune function).
- *Similarity is not shared ancestry:* a typical pair of proteins already has cosine 0.72, so a top match at 0.85 to 0.98 is a ranking. Some matches (SHANK2 → E2F7, APOB → AFP) likely reflect protein size and makeup rather than shared biology.

Two targets for Module 4 (structure-based pocket assessment):

1. *ITGAX (final rank 4, p_ab 0.59):* CD11c marks the immune cells that gather around AD damage. Its relative ITGAL is drugged by lifitegrast in the same kind of ligand-binding domain, which also has a second pocket that experimental drugs bind. Module 4 should test whether ITGAX has similar pockets, which would open a small-molecule route alongside antibodies.
2. *APLNR (final rank 6, p_sm 0.68):* my strongest small-molecule prior, a GPCR with a defined ligand pocket and experimental structures to check the prediction against.

I did not pick PECAM1, despite its first place: it is an antibody target with flat binding surfaces, so pocket assessment would add little. APOE is an exploratory third choice, because ApoE4 has a known site where shape-correcting drugs bind.

Unmapped genes: none of my top 15. All have a reviewed UniProt protein. In my full list of 1,115 genes, the only one without a protein is UBBP4, a ubiquitin pseudogene. A protein model cannot score it; it would need RNA evidence (is it made into RNA in AD brain?), a test of whether its RNA has a role of its own, and genetic evidence linking its expression to AD risk.